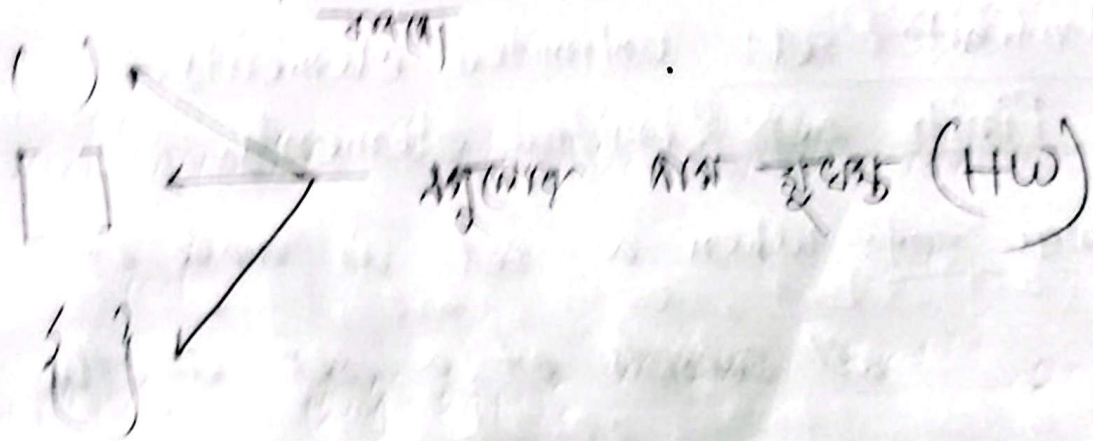


SIT

Definition: Set is an unordered collection of objects / things, using any source factor or



Equality: When two or multiple set have same elements and same number of elements, they are equal sets.

Ex:

$$\{a, b, c\} = \{b, a, c\}$$

$$\{a, b, c\} \neq \{a, b\}$$

$$\{a, b, c\} \neq \{a, a, t\}$$

Subset: When a set is made using elements of another set is a subset.

Empty set: $\{ \}$, $\emptyset$.

Universality: All systems are included
→ A set is universal set

Infinite set: unlimited elements.

Finite set: Limited elements.

Power set: When a set is made of all subsets of a set is the power set.

~~P(A)~~

$$A = \{a, b\}$$

$$P(A) = \{\{a, b\}, \{a\}, \{b\}, \emptyset\}$$

Complement: Complement set, $\bar{A}$

$$\bar{A} = U - A$$

(*) What is the power set of $\{0, 1, 2\}$?

⇒ Suppose,

$$A = \{0, 1, 2\}$$

∴ The subsets are $\{0, 1, 2\}, \{0, 1\}, \{0, 2\},$

$$\{1, 2\}, \{0\}, \{1\}, \{2\}, \{\}$$

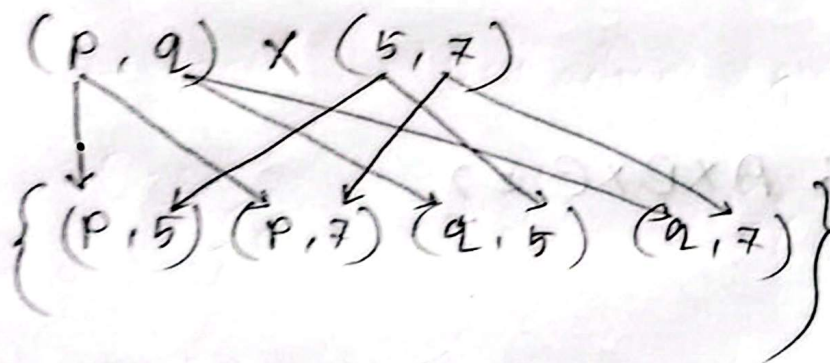
$$\therefore P(A) = \{\{0, 1, 2\}, \{1, 2\}, \{0, 1\}, \{0, 2\}, \{0\}, \{1\}, \{2\}, \{\}\}$$

(A)

(*) Subset of $\emptyset$] special case.

$$= \{\{\emptyset\}, \emptyset\}$$

Cartesian Product:



(*) $A = \{1, 2, 3\}$

; $A \times B = ?$

$B = \{a, b, c\}$

$B \times A = ?$

$\Rightarrow A \times B$
 $= \{1, 2, 3\} \times \{a, b, c\}$

$$= \{(1, a), (2, a), (3, a), (1, b), (2, b), (3, b), (1, c), (2, c), (3, c)\} \quad (A \times B)$$

$$\therefore B \times A$$

$$= \{a, b, c\} \times \{1, 2, 3\}$$

$$= \{(a, 1), (b, 1), (c, 1), (a, 2), (b, 2), (c, 2), (a, 3), (b, 3), (c, 3)\} \quad (B \times A)$$

$$(A \times B) \neq (B \times A)$$

HW:

$$(*) A = \{0, 1\}$$

$$B = \{1, 2\} ; A \times B \times C = ?$$

$$C = \{0, 1, 2\}$$